AMAN SHARMA

Software Engineer — Backend · Systems · Data · Embedded

Jammu, India • +91-7006551826 • official.aman1605@gmail.com

linkedin.com/in/amansharma16050 • github.com/sharma16aman

Professional Summary

Software Engineer and M.Tech graduate (NIT Jalandhar) with hands-on industry experience at Intel Corporation building production-grade ETL pipelines, REST API integrations, and CI/CD-driven validation automation for GPU software. Published researcher with a Springer conference paper on ML-driven IoT edge intelligence. Proficient in Python, C, C++, and system-level programming, with demonstrated ability to deliver end-to-end solutions across backend engineering, data engineering, embedded systems, and applied machine learning.

Experience

Software Engineering Intern — GPU Software Validation **Jul 2025 – June 2026**

Intel Corporation *Bangalore, India*

Gaming Data Pipelines

- Engineered an end-to-end ETL pipeline monitoring **30–50 active game titles per release cycle** (from a pool of 300–400); auto-refreshed driver health data every **2 hours** via Salesforce API and surfaced live per-game, per-feature readiness through Power BI dashboards for Day-0 pre-launch decisions.
- Delivered a CLI batch ingestion tool in **under 2 days** with Excel input, built-in data validation, and log/folder auto-parsing — rejecting dirty/malformed records at ingestion to keep the database clean, while cutting manual entry effort by **70–80%**.
- Automated driver scorecard generation across **300+ game titles**, **10+ platforms, and multiple SKUs** using a JSON feature-map for code-free feature onboarding; enabled direct Intel GPU vs. competitor comparisons across the full supported hardware portfolio.

Gaming Performance & Graphics Metrics

- Computed **110+ functional and performance metrics** (avg FPS, percentile frame times, stutter KPIs, and feature-specific indicators) from raw PresentMon benchmark logs across **250+ game titles**; synced validated results to the database and test plan via REST API.
- Extended GPU benchmark tooling with selectable FPS modes for AI frame-generation titles (XeFG/MFG), improving stutter reporting accuracy and resolving a data-limit bug in the report-generation API.
- Built a directory-traversal scanner across **300+ game builds** to detect Intel XeSS, XeFG, and FG feature libraries, extract version metadata, and automatically flag missing or incorrect integrations before release cycles.

Test Plan Automation & Validation

- Designed a REST API-driven sync system reconciling enabled/disabled/blocked states across **1,000+ test items** spanning multiple manual and automated validation plans — eliminating all duplicate manual mirroring at the start of each release cycle.
- Developed scheduled GitHub Actions monitors with configurable frequencies (every 2 hours to daily) to detect concurrent state changes across shared test plans and auto-dispatch CSV-based email alerts to validation stakeholders.
- Built an audit script continuously validating test plan integrity as new items were added; generated multi-sheet Excel reports for validation leads, reducing misconfiguration and coverage gaps across active plans.

Database Design & Workflow Automation

- Designed relational schemas across **4 domains** (Day-0, benchmark, functional, performance); managed staging-to-production migrations and scheduling tables powering **8 active ETL pipelines**.
- Implemented a centralized scheduling table with pipeline signatures and active-state filtering; reduced ETL session management from **2+ weeks of manual DB work to under 2 days** via CLI — eliminating the need for direct database access entirely.
- Configured a self-hosted GitHub Actions runner as a persistent system service, executing **50+ automated workflows daily** with reliable 24/7 uptime for all scheduled validation pipelines.

Research Publication

ML-Driven Adaptive Edge-to-Cloud Framework for Intelligent IoT Data Reduction and Strategy Optimization

Aman Sharma (Sole Author)

2nd International Conference on Power and Intelligent Control Systems (PICS-2026), NIT Hamirpur — Springer Proceedings

Status: Presented successfully; awaiting DOI assignment.

- Proposed a hybrid Transformer-BiLSTM multi-task learning model for dynamic IoT reduction strategy selection and adaptive threshold prediction, achieving 98.74% data reduction ratio and RMSE of 0.041 on synthetic IoT streams.
- Evaluated the framework on the Intel Berkeley Research Lab dataset (2.3M+ readings, 54 sensors), achieving 67.83%

reduction with RMSE of 0.063, outperforming all static and ML-based baselines.
- Attained 96.12% strategy-selection accuracy and 95.48% F1-score; CBOR serialization reduced message size by 56.4% over JSON without affecting reconstruction quality.

Projects

M.Tech Thesis: Unified Edge-to-Cloud Framework for Smart IoT Data Reduction *2024 – 2026*
Tech Stack: Python, TensorFlow, Transformer, BiLSTM, MQTT, HTTP, JSON, CBOR, Pandas

- Designed and implemented a two-stage framework: (1) a unified evaluation benchmark jointly testing 4 data-reduction strategies, two communication protocols, and two serialization formats under identical conditions; (2) an ML-driven adaptive framework using a hybrid Transformer-BiLSTM model for real-time strategy selection and threshold regression.
- Demonstrated chained reduction strategies achieving up to 99.14% data reduction with RMSE ≤ 0.105 on synthetic data; validated on real-world Intel Lab IoT dataset revealing a critical synthetic-to-real performance gap and bridging it with the adaptive ML framework.
- Implemented full edge-to-cloud pipeline including data preprocessing, temporal feature engineering, multi-task learning, MQTT/HTTP communication, and CBOR/JSON serialization benchmarking.

User-Space Device Driver Simulation and Validation Framework *2026*
Tech Stack: C, Linux, Pthreads, Makefile

- Built a user-space driver simulation in C with hardware-style memory-mapped registers, interrupt-driven communication, and a thread-safe ring-buffer data transfer mechanism using POSIX threads.
- Implemented DMA-like bulk transfer, fault injection, automated regression tests, structured logging, and performance metrics for validation-focused embedded systems debugging.

Real-Time Driver Drowsiness Detection System *2024*
Tech Stack: Python, TensorFlow, OpenCV, CNNs

- Developed a real-time driver drowsiness detection system using convolutional neural networks and computer vision with facial landmark feature extraction; achieved 92% classification accuracy.

Technical Skills

Languages: Python, C, C++
Backend & APIs: REST API design, API integration, CLI tooling, batch processing, microservices, automated reporting
Systems & Embedded: Linux, POSIX threads (Pthreads), memory-mapped I/O, interrupt handling, DMA, ring buffers, Makefile
Data Engineering & ETL: Multi-stage ETL pipelines, Salesforce API, PresentMon, pandas, JSON-driven config systems, data modeling
Databases: MySQL, MariaDB, MongoDB — schema design, staging-to-production migrations, multi-domain DB management
Machine Learning: TensorFlow, Transformer, BiLSTM, CNNs, multi-task learning, time-series modeling, MQTT, CBOR
CI/CD & DevOps: GitHub Actions, self-hosted runners, scheduled/manual pipeline orchestration, Git
Tools & Reporting: Excel automation (openpyxl), Power BI, multi-sheet report generation, audit scripting
Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, System Design

Education

Master of Technology (M.Tech) — Computer Science & Engineering **2024 – 2026**
Dr. B.R. Ambedkar National Institute of Technology Jalandhar *Jalandhar, India* CGPA: 8.23/10
Thesis: A Unified Edge-to-Cloud Framework for Smart IoT Data Reduction

Bachelor of Engineering (B.E.) — Computer Science & Engineering **2018 – 2022**
MIET, University of Jammu *Jammu, India* CGPA: 8.71/10

Certifications & Achievements

- **GATE 2024** — Qualified (Graduate Aptitude Test in Engineering, Computer Science)
- **NPTEL Certifications:** Joy of Computing using Python · Cloud Computing · Blockchain and its Applications
- **Second Runner-Up** — Jammu & Kashmir CSI (Computer Society of India) Convention, 2021
- **Secretary** — ACM Student Chapter, MIET Jammu (2021)